00-W-891c
August 25, 1967
SUPERSEDING
Int. Fed. Spec. 00-W-00891d(GSA-FSS)
May 13, 1965 and
Fed. Spec. 00-W-891c
December 19, 1963

FEDERAL SPECIFICATION

WRENCH, IMPACT, PNEUMATIC, PORTABLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable pneumatic impact wrenches, for tightening and loosening bolts and nuts, that are most generally used by the Federal Government.

1.2 Classification.

1.2.1 Sizes. The wrenches shall be furnished with the following square male drive sizes, as specified (see 3.11 and 6.1):

Size 3/8 (1/4-inch capacity).
Size 1/2 (3/8-inch capacity).
Size 5/8 (5/8-inch capacity).
Size 3/4 (3/4-inch capacity).
Size 1 (1-1/4-inch capacity).
Size 1-1/2 (1-3/4-inch capacity).

2. APPLICABLE DOCUMENTS

2.1 Specification and standards. The following specifications and standards, of issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:
MIL-P-12829 - Packaging of Pneumatic Hand Tools with Attachments and Accessories.

Military Standard:
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:
H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.)

United States of America Standards Institute (USASI) Publication:
B5.38 - 1958 - Driving and Spindle Ends for Portable Air and Electric Tools.

(Copies may be obtained from the United States of America Standards Institute, 10 East 40th Street, New York, N.Y., 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are descriptive, and not restrictive, and are not intended to preclude the purchase of wrenches otherwise conforming to this specification.

3.2 Qualification. Portable pneumatic impact wrenches furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on, or approved for listing on the applicable qualified products list.

3.3 Material. Material used in the construction of the wrenches shall be of a good commercial quality suitable for the intended purpose. All materials used shall be free from defects and imperfections that may adversely affect the serviceability of the finished product.

3.4 Screw threads. All threaded parts, including screws, bolts, and nuts, shall conform to the applicable requirements of NBS H28.

3.5 Identification of product. Wrenches shall be marked in a plain and permanent manner, such that the source of manufacture can be easily determined.

3.6 Lubrication.

3.6.1 Primary lubrication. Wrenches shall be provided with an oil reservoir. The oil reservoir shall be of sufficient capacity so that one filling will supply lubrication to moving parts in the path of the air stream for at least 8 hours of continuous operation. The oil shall be fed from the reservoir by an automatic device that functions through an action dependent upon the operation of the wrench. The oil flow shall be regulated so that lubrication will be sufficient without excessive quantities of lubricant being discharged with the exhaust air.

3.6.2 Secondary lubrication. Conventional type lubricating fittings or openings or other suitable means shall be provided for introduction of sufficient lubricant for bearings, gears, and other moving parts not lubricated from the oil reservoir. The fittings and openings shall be such that lubrication is required only once in not less than 48 hours of actual operation. Where acceptable prelubricated sealed bearings are used, no external means of lubrication shall be required for these parts.

3.7 Air inlet connection. The air inlet connection shall consist of a well-secured ferrous fitting threaded with internal American Standard taper pipe threads (NPT) or internal Dryseal American Standard taper pipe threads (NPTF) in sizes specified in table I. The air inlet shall be fitted with an air strainer or screen (20 mesh or finer) which shall be effective in retaining solid particles in the compressed air supply. The strainer or screen shall be of such design that it can be easily removed for cleaning.

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3.8 Bearings. All rotating parts of the wrenches shall be provided with commercially available bearings. The rotor shaft shall be provided with ball bearings. Needle and roller bearings, which are primarily designed for radial loads, shall not be used in a thrust position. If open type bearings are used, they shall be suitably housed and adequately sealed to prevent leakage of lubricant and entrance of dirt and dust. The drive end of the spindle may be supported by a suitable inserted sliding contact type bearing (sleeve bearing).

3.9 General construction. Wrenches shall be of rugged construction so as to withstand, without failure, treatment likely to be encountered under general service conditions. Wrenches shall be self-contained, readily accessible for adjustment or replacement of parts. Parts which require lubrication shall be properly enclosed so as to prevent entrance of foreign particles and leakage of lubricant. Working parts exposed to wear shall be of sufficient hardness to withstand the service required. Parts shall be properly finished by machining or grinding to dimensions within limits established by good commercial practice. Similar parts shall be interchangeable and of good fit when assembled with relative parts of the same size, type, and make. Wrenches shall be reliable and effective when operated on compressed air lines at gage pressures between 80 and 100 pounds per square inch (p.s.i.).

3.10 Housing. Housing shall be strong enough to prevent indentation under service conditions. When threaded fasteners are utilized to assemble adjacent housing components, through bolting shall be used in preference to tapped holes. Cap screws, if used, shall engage threads in ferrous housings or ferrous inserts in nonferrous housings by not less than a length equal to the pitch diameter of the fastener thread. Where ferrous inserts are not used in nonferrous housings, cap screws shall engage threads by not less than length equal to 1-1/2 times the pitch diameter of the fastener thread. All threaded fasteners shall be secured with suitable locking devices.

3.11 Spindle. The drive end of the spindle shall be wear-resistant. The drive end, when fitted with a socket, shall not show excessive run-out which may adversely affect operation of the wrench. The drive end of the spindle shall conform to USASI B5.38 for the nominal square male drive size, as specified for the applicable size wrench in table I.

3.12 Operation. The wrenches shall be reversible. The rotational direction of the wrench shall be controlled by a conveniently located mechanical device which is independent of the throttle. The rotational control shall be plainly and permanently marked to indicate direction of rotation.

3.13 Impact unit. Wrenches shall contain an impact unit which is actuated by the motor. The impact unit shall be designed to convert torque from the motor into rotary impacts and to transmit these impacts to the spindle.

3.14 Sizes. The wrenches shall conform to the requirements of table I for the size specified (see 6.1), and shall be similar to figures 1, 2, 3, 4, or 5, as applicable. The performance criteria of table I applies to the wrenches in the forward(counterclockwise rotation of the spindle looking at the front of the wrench) direction only:

TABLE I. Weight, dimensional and performance requirements

Spindle size	Weight max.	Length max.	Spindle offset max.	Air inlet (NPT-NPTF)	Air consumption 1/ at load max.	Time to twist bolt specimen 180° max.
Inches	Pounds	Inches	Inches	Inch	CFM	Seconds
3/8	5	9-1/4	1-1/4	1/4	15	5
1/2	7	10-5/8	1-3/8	1/4	30	5
5/8	12	12	1-5/8	1/4	35	5
3/4	19	16	2	3/8	50	5
1	35	21-1/8	2-1/2	3/8	60	5
1-1/2	73	25-5/8	3	1/2	85	10

1/ Air consumption at load measured during test specimen twist test.

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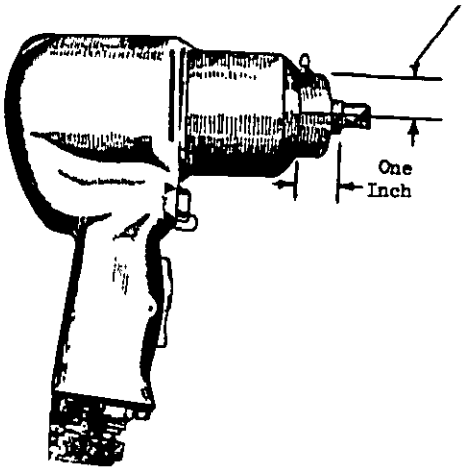


FIGURE 1. Pistol style handle.

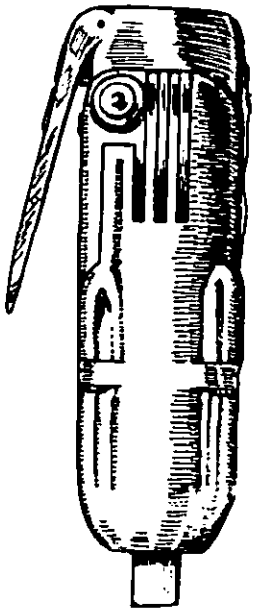


FIGURE 2. Body grip style handle.

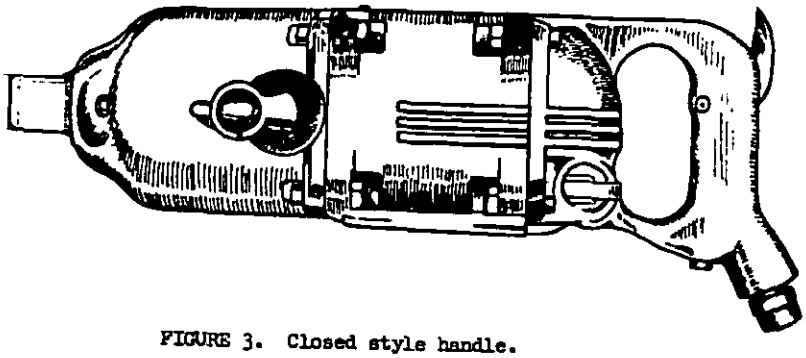


FIGURE 3. Closed style handle.

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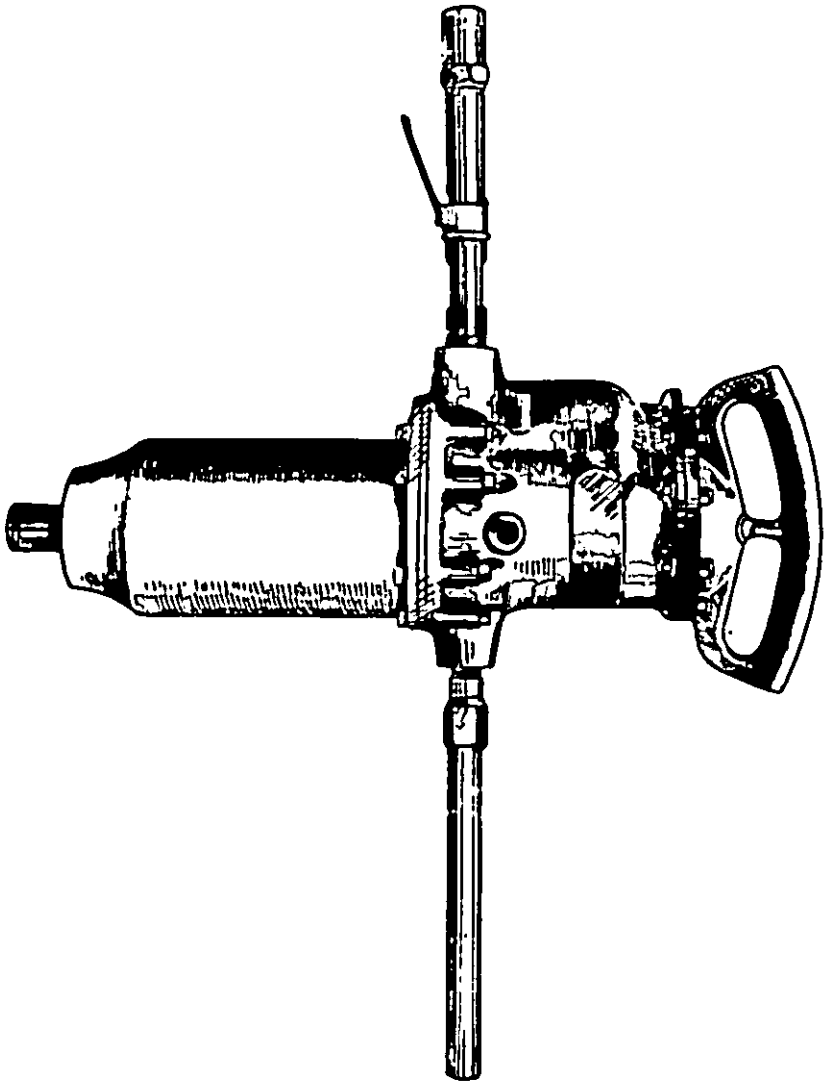


FIGURE 4. Fixed handle or grip.

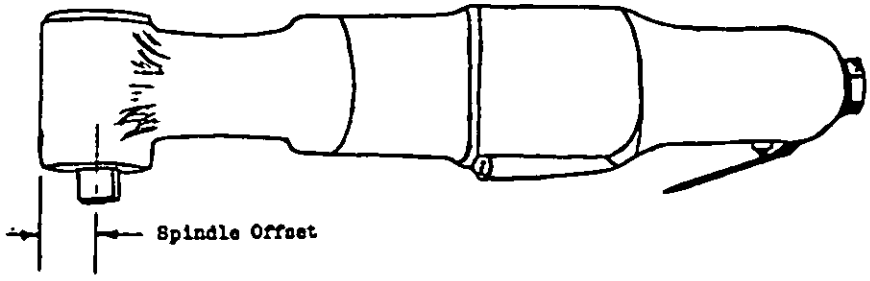


FIGURE 5. Body grip style handle, angle drive.

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3.15 Angle drive. When specified (see 6.1), the 1/4 and 3/8 inch wrenches shall be of the angle drive type with the axis of the drive shaft 90° from the axis of the wrench housing.

3.16 Handles.

3.16.1 Sizes 3/8 and 1/2 inch. Each wrench of the 3/8 and 1/2 inch size shall be equipped with a pistol-style handle (fig. 1) or a body grip-style handle figure 2 or 5 (when applicable), as specified (see 6.1). The pistol-style handle tool shall be fitted with the air-inlet connection at the base of the handle. The body-grip style handle tool shall be fitted with the air-inlet connection at the end of the housing opposite the spindle.

3.16.2 Sizes 5/8 to 1 inch. Each wrench of the 5/8 to 1 inch size shall be equipped with a closed-style handle (fig. 3) located at the end of the housing opposite the spindle or a pistol-style handle (fig. 1), as specified (see 6.1). Sizes 5/8 to 1 inch wrenches shall be fitted with the air-inlet connection at the base of the handle. Wrenches furnished with closed-style handles shall be equipped with a removable "dead" handle located in or on the housing, perpendicular to the spindle axis and to the plane of the closed-style handle.

3.16.3 Size 1-1/2 inch. Each wrench of the 1-1/2 inch size shall be equipped with a fixed handle or grip (fig. 4) located at the end of the housing opposite the spindle, a removable "dead" handle located in the side of the housing perpendicular to the spindle axis, and a "live" handle in the side of the housing in alignment with the "dead" handle. The "live" handle shall be fitted with the air-inlet connection.

3.17 Throttle. Wrenches shall be provided with a nonleakable manual throttle for controlling the supply of compressed air. Throttles shall be of such design that the speed can be regulated during the operation. Wrenches of the 1/4 to 1 inch sizes shall be equipped with a trigger, button, or lever style throttle on the handle. Wrenches of the 1-1/2 inch size shall be equipped with a sleeve or lever style throttle on the "live" handle.

3.18 Accessories or extra equipment. When specified (see 6.1), accessories or extra equipment shall be furnished.

3.19 Air volume (torque) regulator. The wrenches shall be furnished with an air volume (torque) regulator. The regulator may either be built into the wrench or furnished separately. It shall have varying degrees of restriction for regulating the volume of air flow. When in the wide open position, the wrench air consumption shall not exceed that shown in table I.

3.20 Instruction book. One copy of an instruction book consisting of the manufacturer's standard commercial instructions and parts list, bound together, shall be furnished with each wrench.

3.21 Workmanship. The wrenches and accessories shall be free from all imperfections which will adversely affect the general appearance, function, or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or if none, in accordance with this specification.

4.2 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center and shall consist of the tests specified in 4.6 and 4.7, and the examination specified in 4.4. Tests shall be conducted on each size impact wrench for which qualification is desired. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

4.3 Quality conformance inspection.

4.3.1 Inspection lot. Wrenches of the same size shall be considered a lot for the purpose of quality conformance inspection. Inspection shall include the examination of 4.4, the test of 4.6, and the inspection for delivery of 4.5.

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4.3.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table II.

TABLE II. Sampling data

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	One wrench	II	1.5	Percent defective	4.4.1
Dimensional and weight examination	One wrench	II	1.5	Percent defective	4.4.2
Testing for quality conformance	One wrench	II	1.5	Percent defective	4.6
Preparation for delivery	One wrench packaged for delivery	II	4.0	Percent defective	4.5

4.4 Examination.

4.4.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, construction, workmanship, and marking. Defects are listed in table III.

TABLE III. Classification of defects

No.	Defects
1	Material nonconforming.
2	Wrench not easily adjustable or parts not easily replaceable.
3	Lubricated parts enclosure nonconforming.
4	Similar parts not dimensionally and functionally interchangeable.
5	Air-inlet connection not as specified, damaged, or missing.
6	Air strainer missing or damaged, not easily removable.
7	Marking, nonconforming, illegible, incorrect, or missing.
8	Handles not as specified or damaged, not removable, when required.
9	Throttle nonconforming, damaged, inoperable, or missing.
10	Spindle not as specified.
11	Bearings not as specified, broken, cracked, or pitted.
12	Lubrication of wrench not as specified; evidence of oil leakage.

4.4.2 Dimensional and weight examination. Each sample unit shall be examined for any nonconformance with dimensional and weight requirements.

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined for conformance with the requirements of section 5.

4.6 Test specimen twist test. Sample wrenches shall be subjected to a test to determine compliance with the twist time requirements specified in table I, for the applicable size wrench being tested.

4.6.1 Apparatus. The test set-up shall consist of a fixture for holding the test specimen, similar to that shown in figure 6. The specimen shall be machined from annealed, unfinished hexagon bar stock, conforming to SAE grade 1 commercial steel, having a minimum tensile strength of 55,000 p.s.i., containing not more than 0.05 percent phosphorus and 0.06 percent sulphur, with a maximum hardness of 95 on the Rockwell B scale. The specimen shall be of the shape shown in figure 7 and the dimensions specified in table IV for the size wrench being tested.

TABLE IV. Dimensions of test specimens

Size of wrench, inches	Nominal 1/ hex, size A, inches	Overall length B, inches	"C" length, inches, ±0.005	"D" diameter, inch, ± 0.0005	"E" diameter, inch, ± 0.005
1/4	1/2	4	0.500	0.250	0.480
3/8	5/8	4-1/4	.750	.320	.680
5/8	7/8	4-1/2	1.000	.430	.680
3/4	1	4-1/2	1.000	.510	.880
1-1/4	1-1/8	4-1/2	1.000	.680	.980
1-3/4	1-3/8	5	1.500	1.000	1.000

1/ Distance across flats of hexagon bar.

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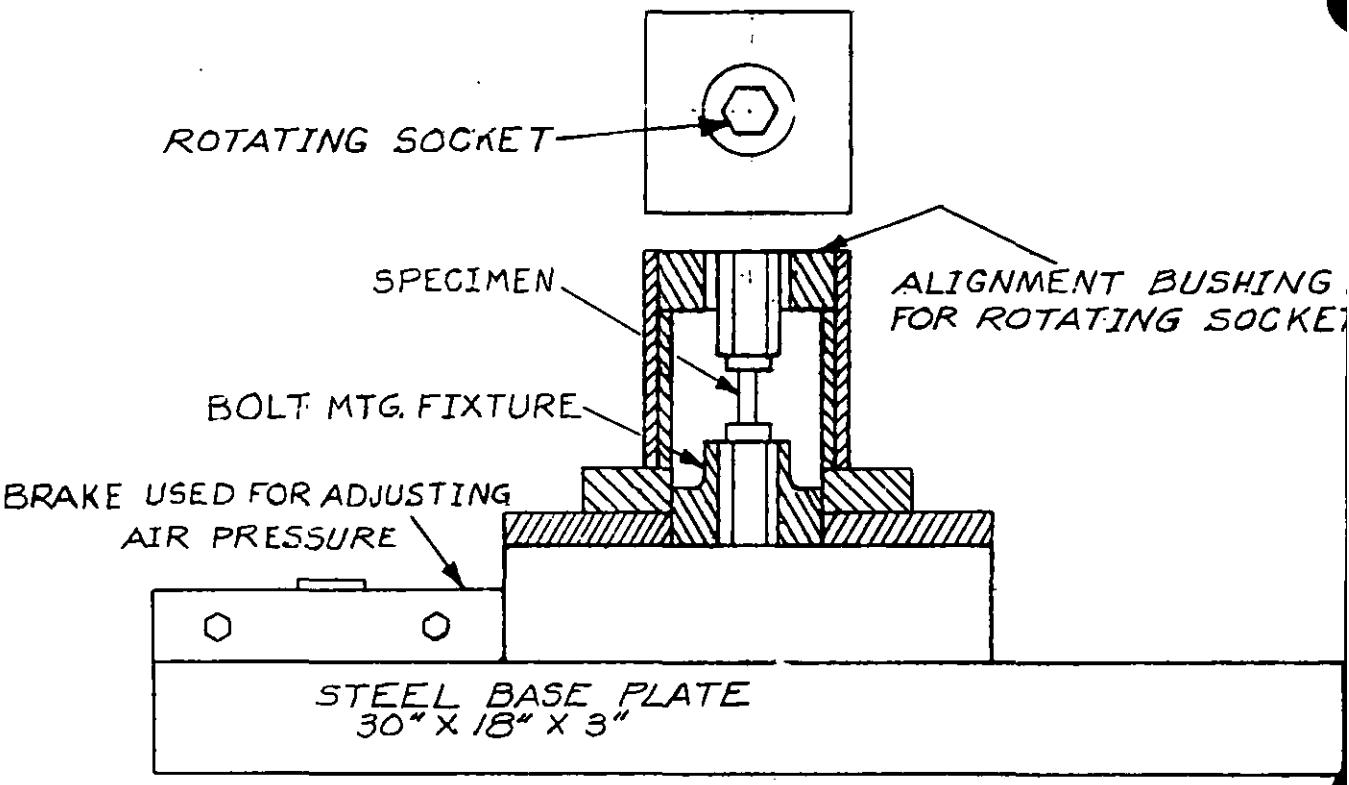


FIGURE 6. Cross sectional drawing of "Time to Twist" set-up for pneumatic wrenches

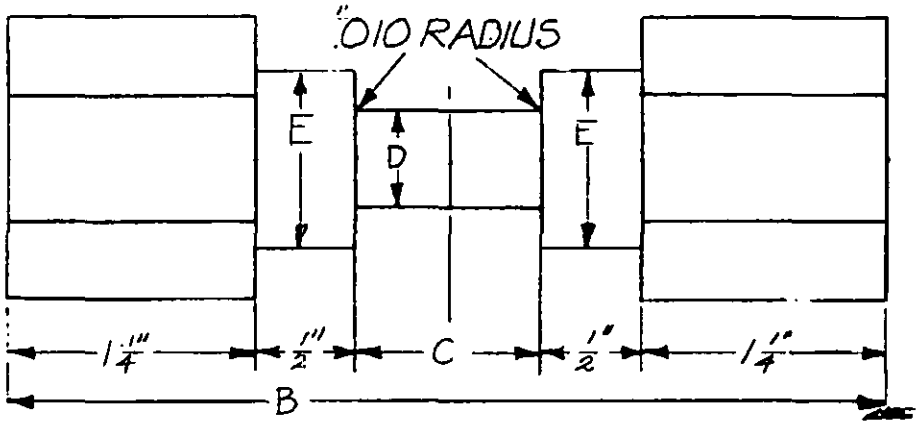


FIGURE 7. Hexagon bar twist test specimen.

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4.6.2 Procedure. A test specimen of the applicable size shall be mounted in the test fixture. The air supply to the wrench shall be adjusted by a means of a pressure regulator to a constant air pressure of 90 p.s.i. while the tang is rotating at approximately 15 revolutions per minute. The wrench shall then be mounted on the test specimen, using a socket of the same hexagon size as the hexagon size of the test specimen. The wrench shall be allowed to impact until the specimen twists 90°. Using this initial 90° twist as a datum point, the wrench shall again be allowed to impact for the time specified for the applicable size wrench being tested (see table I), and the test specimen shall have twisted an additional 180°. The time required to twist the specimen and the air consumption, while impacting, shall be recorded. Three runs shall be made with each size wrench being tested and the average values obtained for air consumption and time to twist the bolt specimen. Average values exceeding those specified in table I shall be cause for rejection. If a bolt specimen breaks prior to being twisted 180°, it shall be discarded and a new specimen substituted.

4.7 Service test (qualification testing only). Each size wrench submitted for qualification testing (see 4.2) shall complete 150 hours of actual work under practical service conditions. Each shall first be inspected in accordance with 4.4 and shall be tested in accordance with 4.6. The wrench shall then be put into field use to determine the ease of operation, general serviceability, suitability, and durability of the tool. Failure during the service test of any components of the tool, exclusive of attachments, shall be cause for rejection. After the service test has been completed, the wrench shall be completely disassembled and all parts examined. Any part broken, cracked, or deformed, or not in accordance with the requirements of this specification, shall be cause for rejection. Wrenches completing the service test and disassembly examination shall be reassembled and once again subjected to the test of 4.6.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Impact wrenches shall be preserved and packaged in accordance with MIL-P-12829A, level A or C, as specified (see 6.1). Packing shall be in accordance with MIL-P-12829A, level A, B, or C, as specified (see 6.1).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order (see 6.1), interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special marking required by the contract or order (see 6.1), interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2.1 and 3.14, table I).
(c) Whether wrenches of the 3/8- and 1/2-inch size shall be of the angle drive type (see 3.15).
(d) Handle style required, as applicable (see 3.16).
(e) Accessories or extra equipment, when required (see 3.18).
(f) Applicable preservation, packaging, and packing requirements (see 5.1).
(g) Special marking, if required (see 5.2).

6.2 Definitions.

6.2.1 Spindle offset. The spindle offset, specified in table I, is defined as the distance from the center line of the spindle to the edge of the wrench housing, measured one inch from the base of the spindle on the housing (for example see fig. 1). For the angle drive type, see figure 5.

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6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion on Qualified Products List (QPL) 00-W-891, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Naval Ship Engineering Center, Washington, D. C., 20360, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with Provisions Governing Qualification. (Copies may be obtained from the Commanding Officer, Naval Supply Depot, 5801 Tabor Ave., Philadelphia, Pa., 19120.)

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